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USING JAVASCRIPT

and the Document Object Model

OVERVIEW

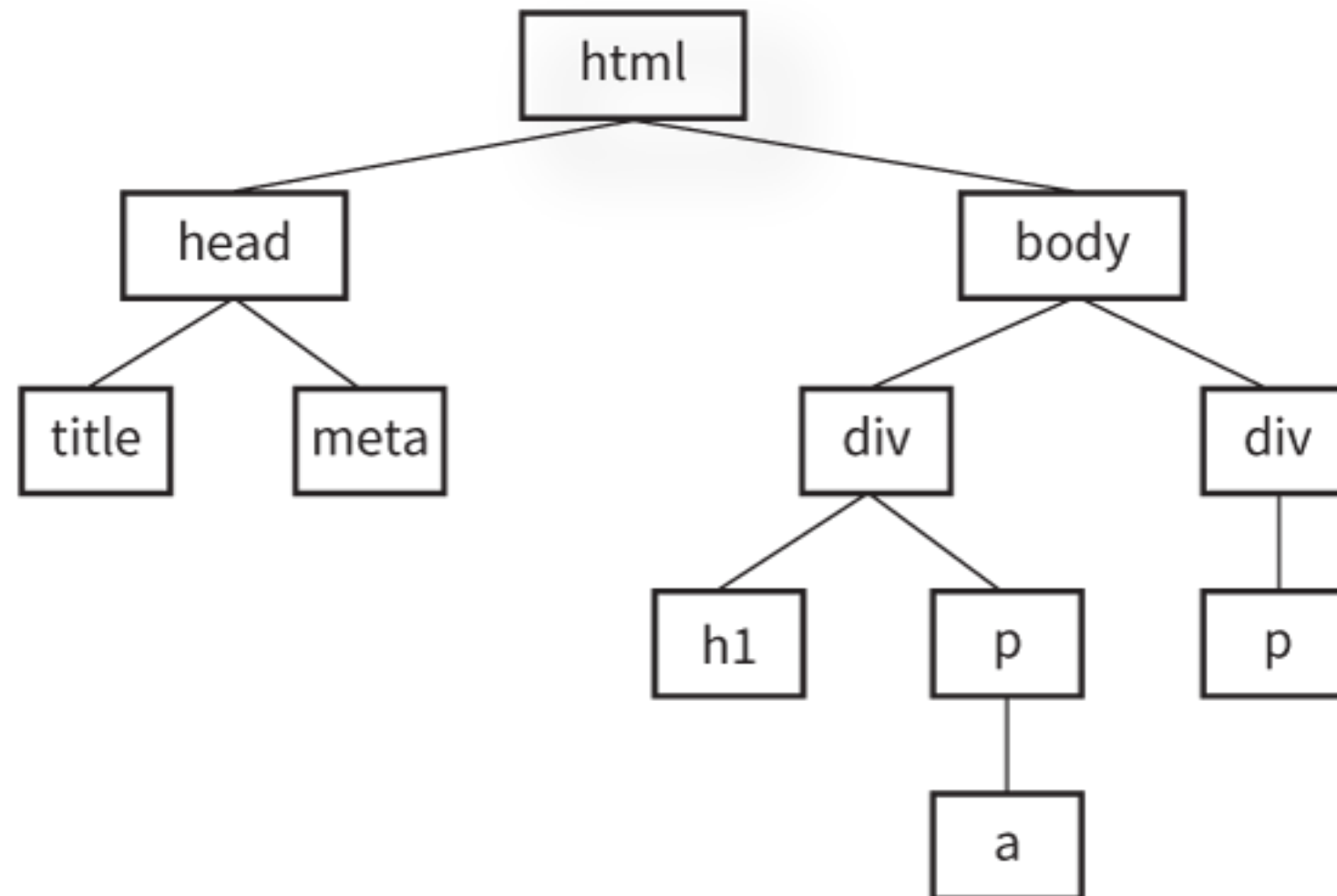
- **What the DOM is**
- **Accessing and changing elements, attributes, and contents**
- **Polyfills**
- **JavaScript libraries**

Intro to the DOM

- The **Document Object Model (DOM)** is a **programming interface** that provides a way to access and manipulate the contents of a document.
- It provides a structured **map of the document** and a set of **methods** for interacting with them.
- It can be used with other XML languages and it can be accessed by other programming languages (like PHP, Ruby, etc.).

Node Tree

The DOM treats the structure of a document like a tree with branches:

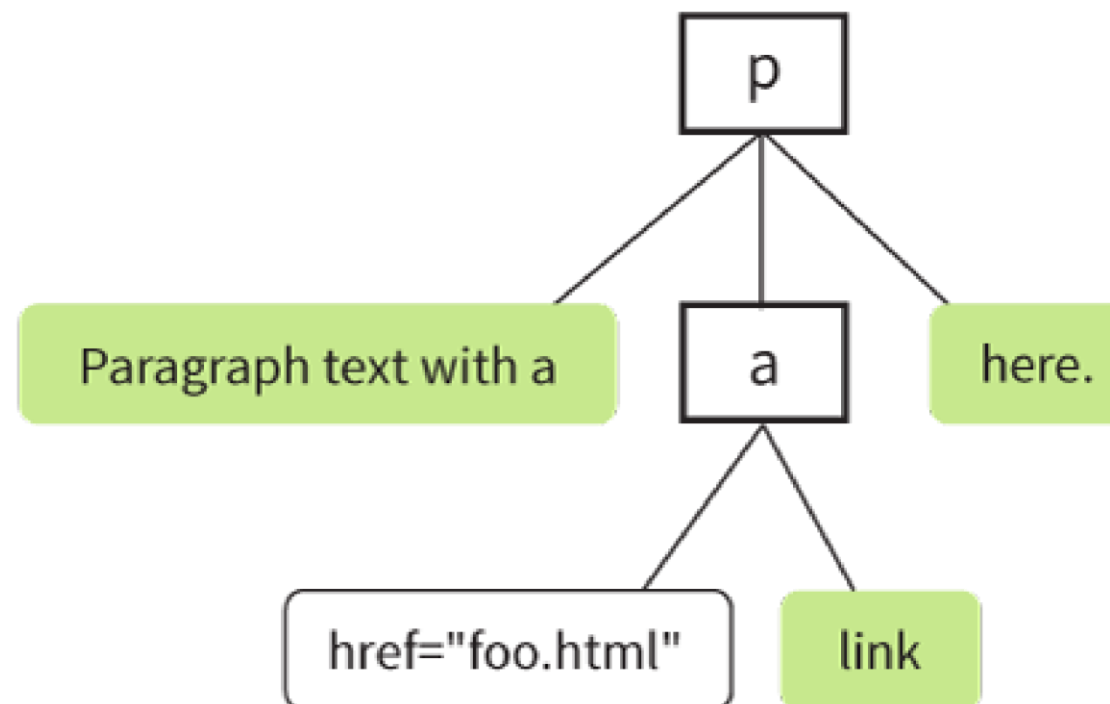


Node Tree (cont'd)

Every element, attribute, and piece of content is a node on the tree and can be accessed for scripting:

The nodes
within a `p`
element

```
<p>Paragraph text with a <a href="foo.html">link</a> here.</p>
```



Accessing Nodes

To point to nodes, list them separated by periods (.).

In this example, the variable `foo` is set to the HTML content of an element with `id="beginner"`:

```
var foo = document.getElementById("beginner").innerHTML;
```

- The `document` object points to the page itself.
- `getElementById` specifies an element with the `id` "beginner".
- `innerHTML` stands for the HTML content within that element.

Accessing Nodes (cont'd)

Methods for accessing nodes in the document:

`getElementsByTagName()`

Accesses all elements with the given tag name

Example: `document.getElementsByTagName("p");`

`getElementById()`

Accesses a single element by the value of its `id` attribute

Example: `document.getElementById("special");`

`getElementsByClassName()`

Access elements by the value of a class attribute

Example: `document.getElementsByClassName("product");`

Accessing Nodes (cont'd.)

`querySelectorAll()`

Accesses nodes based on a CSS selector

Example: `document.querySelectorAll(".sidebar p");`

`getAttribute()`

Accesses the value of a given attribute

Example: `getAttribute("src")`

Manipulating Nodes

There are several built-in methods for manipulating nodes:

`setAttribute()`

Sets the value of a given attribute:

```
bigImage.setAttribute("src", "newimage.jpg");
```

`innerHTML`

Specifies the content inside an element (including markup if needed):

```
introDiv.innerHTML = "<p>This is the intro text.</p>";
```

`style`

Applies a style using CSS properties:

```
document.getElementById("intro").style.backgroundColor = "#000";
```

Polyfills

A **polyfill** uses JavaScript to make new features work in browsers that don't natively support them.

- **Picturefill:** Enables support for `picture`, `srcset`, and `sizes`
 - **Selectivizr***: Allows IE 6–8 to support CSS3 selectors
 - **HTML5 shiv***: Allows IE6–8 to recognize HTML5 elements
- *If you don't need to support IE 8 and earlier, you don't need these polyfills.

JavaScript Libraries

- A **JavaScript library** is a collection of prewritten functions and methods that you can use in your scripts to accomplish common tasks or simplify complex ones.
- Some are large frameworks for building complex applications.
- Some are targeted to specific tasks, such as forms or math.
- The most popular library is **jQuery**.
- Try searching “JavaScript library for _____” to see if there are pre-made scripts you can use or adapt to your needs.